Multiscale Feature Enhancement Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Salient object detection (SOD) in optical remote sensing images (RSIs) is a valuable and challenging task. Although many SOD methods for RSIs have been proposed, there are still some problems, such as insufficient feature extraction, relatively worse detection results for tiny objects, and serious interference by background clutter and lighting shadows. In this article, we propose a novel multiscale feature enhancement network (MFENet) for SOD in optical RSIs. Specifically, MFENet first uses the global feature perception (GFP) module to extract the multiscale and global feature information of salient object regions. Second, the global features are enhanced from the channel and spatial dimensions by using the mixed feature enhancement (MFE) module. Third, the semantic feature guidance (SFG) module is constructed to guide the model to obtain fine-grained semantic feature information. Finally, the boundary optimization module (BOM) is designed to refine the boundary features and obtain accurate saliency detection results. To evaluate the performance of MFENet, extensive experiments are conducted on the publicly available ORSSD, EORSSD, and ORSI-4199 datasets. The quantitative and qualitative results show that our method outperforms the existing state-of-the-art SOD methods in several evaluation metrics and exhibits better inference speed (FPS) and calculation parameters. The code and results of MFENet are available at `https://github.com/darkseid-arch/MFENet`.**

***Index Terms*— Boundary optimization, feature enhancement, feature perception, remote sensing images (RSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) is based on human visual attention to select the key objects/regions in the image [1]. The purpose of SOD is to obtain the position and scale information of salient objects and output the saliency value of each pixel [2], [3]. Currently, SOD is widely used in various computer vision tasks, such as image segmentation [4], object recognition [5], image retrieval [6], and image classification [7]. Although SOD is different from object detection and semantic segmentation to solve specific visual tasks, it can be used as an auxiliary component to extract the image foreground region and suppress background information interference. The last decades have witnessed the remarkable progress of SOD for natural scene images (NSIs), while only a limited amount of studies focused on SOD for optical remote sensing images (RSIs). Different from natural images, saliency detection for optical RSIs has the following difficulties: 1) the RSIs cover a wide area so that the image contains different scenarios and object categories; 2) the high-altitude shooting leads to the diversity of object scales and orientations in the optical RSIs; 3) interference by imaging conditions, the optical RSIs contain a lot of noise and background interference; and 4) optical RSIs contain a large number of tiny object categories, which brings challenges to saliency detection. The characteristics of RSIs limit the direct application of existing NSIs saliency detection methods for optical RSIs.

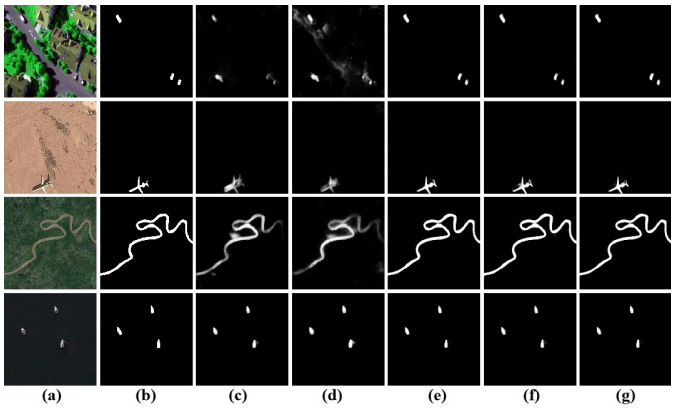


Fig. 1. Visual illustration of SOD results for optical RSIs by applying different methods. (a) Optical RSIs. (b) Ground truth. (c) LVNet [9]. (d) PDFNet [10]. (e) SARNet [11]. (f) DAFNet [12]. (g) MJRBM [13].

Most of the existing optical RSIs SOD methods are based on end-to-end fully convolution neural networks (FCNs) [8],

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which achieve salient object localization and extraction by capturing different feature information of optical RSIs. Although existing FCN-based optical RSIs' SOD methods have achieved better results, there are still some problems. For example, LVNet [9] uses the two-stream pyramid module to obtain the multiscale and spatial details features and the encoder–decoder module is used to suppress the background information. From Fig. 1(c), we observe that LVNet is difficult to obtain the entire salient object region. PDFNet [10] uses the parallel fusion method to effectively fuse the multiscale feature obtained by different convolution modules and uses the dense connections for feature transfers. However, as shown in Fig. 1(d), the saliency detection results of PDFNet for tiny objects are relatively worse. SARNet [11] uses the feature aggregation method to obtain the multiscale and global semantic features and uses global semantic information to guide the model to obtain salient detection results. Fig. 1(e) shows that SARNet cannot accurately obtain the salient object region in complex background. DAFNet [12] uses the context-aware attention module to obtain the semantic context relationship and uses the dense attention fluid structure to generate the global attention feature information. As shown in Fig. 1(f), DFANet obtained the complete salient object region, but the complex background makes it difficult to get the salient boundary. MJRBM [13] uses the hierarchical attention mechanism to obtain the multiscale feature and combines the boundary feature information with the region feature information to obtain salient region detection results. As shown in Fig. 1(g), MJRBM obtains accurate salient detection results, while the detection results for tiny object regions are relatively worse. In summary, the existing optical RSIs SOD methods have problems, such as insufficient global feature extraction, unable to extract the feature information of tiny object regions, ignoring the guidance of semantic feature on global feature information, and poor salient boundary detection result.

To handle the above problems, we propose a novel multiscale feature enhancement network (MFENet) for optical RSIs SOD. The main contributions of this work can be summarized as follows.

1) An end-to-end MFENet is proposed to achieve SOD in RSIs, equipped with the global feature perception (GFP) module, mixed feature enhancement (MFE) module, semantic feature guidance (SFG) module, and boundary optimization module (BOM), which generalize well to varying scenes and object categories.
2) The GFP module is designed based on encoder–decoder structure, and the spatial perception block is introduced to enhance the spatial perception ability; the MFE module is constructed by combining residual spatial and channel attention mechanisms (CAMs) to improve the feature extraction effect for objects in complex scenes.
3) The SFG module is proposed to combine multiscale feature extraction (MSF), progressive feature fusion (PFF), and recurrent attention mechanism (RAM), which can use deep semantic information to guide global features; the BOM is constructed to obtain the fine boundary features to improve the detection effect for salient object boundary region.

II. Related Works

A. Salient Object Detection for NSIs

Currently, the SOD methods for NSIs include the bottom-up method based on handcrafted features and the top-down method based on deep learning. The bottom-up SOD method uses the designed handcrafted feature or prior visual knowledge to describe the attributes of the salient object, which is given as follows. Lei et al. [2] proposed a universal framework for SOD, which uses distance weighting, adaptive binarization, and morphological closing to obtain coarse salient regions and uses super-pixel segmentation to refine the detection results. Jiang et al. [3] proposed a super diffusion model to achieve SOD, which ensembles different proportion feature spaces of salient objects and uses supervised learning to obtain the best ensemble parameters. Yang et al. [14] proposed an edge-guided SOD method, which decomposes the edge feature into nested super-pixels with different scales and then uses color contrast, spatial prior, and edge prior to obtaining the salient detection results. Nguyen et al. [15] proposed an SOD method based on semantic prior, which embeds the semantic information into the SOD process, constricting the SOD results using the trained model. Liu et al. [16] used the closure completeness and closure reliability features of the saliency region to obtain complete feature space information and used the hierarchical calculation method to obtain the salience detection results. Peng et al. [17] proposed an SOD method based on structural matrix decomposition, which uses sparse regularization and Laplacian regularization to obtain the saliency difference between the object and background regions. Zhang et al. [18] conducted the saliency detection based on color and texture curves, which obtained the saliency region subgraphs via affinity propagation clustering and graph-based manifold ranking. To address the oversegmentation problem in SOD, Zhang et al. [19] proposed a graph-based framework for SOD, which uses different graph models to describe attribute information of the salient regions.

Benefiting from the powerful feature extraction and representation ability of convolutional neural networks (CNNs), many top-down SOD methods based on deep learning have been proposed. Hou et al. [20] proposed an SOD model based on FCNs and skip connection, which uses FCNs to extract the multilevel and multiscale features and uses skip connection for feature transfer. To refine the SOD results, Hu et al. [21] used the recurrent neural network (RNN) to refine the convolution features at each layer, and then, the feature aggregation operation is used to suppress the nonsalient area noise interference. To fully use texture information contained in the object region, Liu et al. [22] proposed a pixelwise texture attention network, which uses the attention map and the corresponding weight information to guide saliency detection. Deng et al. [23] used the residual block to establish the relationship between low- and high-level features, and the feature interaction is used to achieve multiscale fusion between

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different features. Zhao and Wu [24] proposed a pyramid feature attention network to enhance high-level texture and low-level spatial features and used the BOM to obtain refined saliency detection results. Zhao et al. [25] proposed an SOD model with edge information guidance, which uses progressive fusion to obtain coarse multiscale features and uses local edge information to refine saliency detection results. Based on the U-shape structure, Liu et al. [26] proposed a real-time saliency detection network, which uses the global guidance module to obtain different feature information of the potential saliency object. To improve the saliency detection quality of boundary region, Qin et al. [27] proposed a boundary-aware network for saliency detection, which uses the boundary encoder–decoder structure and the residual module to obtain the saliency detection results.

B. SOD for Optical RSIs

Due to the complexity and challenge of the scenarios in optical RSIs, compared with the research boom of SOD for NSIs, only a few works focus on the SOD for optical RSIs. Previously, many SOD methods are used as an auxiliary tool for RSIs interpretation, which are given as follows. Chen and Zhang [28] proposed a clustering method based on global color and shape contrast to obtain the salient regions of RSIs and used the salient parts to extract key feature information. Huang et al. [29] proposed an SOD method based on contrast-weighted dictionary learning, which uses positive and negative samples to construct the discriminant dictionary and uses sparse representation and reconstruction errors to obtain the saliency regions. Dong et al. [30] constructed an SOD method using multiscale analysis and Fourier histogram of oriented gradients (HOG) for remote sensing ship detection, which uses multiscale analysis to obtain multiscale features and uses Fourier HOG for ship saliency edge detection. Zhao et al. [31] proposed a subjective saliency model driven by the multicues for airport detection, which uses active learning to obtain salient region features and uses support vector machine (SVM) to optimize the saliency detection results. Zhang et al. [32] proposed a saliency detection method for extracting the critical information of RSIs, which uses the visual saliency method to obtain the saliency region feature information and uses the active contour model to obtain the saliency detection results. Liu et al. [33] proposed an unsupervised saliency detection model based on the color Markov chain, which uses CIE Lab space to construct the Markov chain and generate salient feature maps.

With the wide application of deep learning in computer vision, many RSIs saliency detection methods based on deep learning are proposed. Li et al. [9] proposed an end-to-end saliency detection model LV-Net, which includes the two-stream pyramid module and the encoder–decoder module with nested connections. The LV-Net can fully extract the multiscale feature and spatial structure information contained in RSIs. Li et al. [10] proposed a novel parallel down-up fusion network for saliency detection, which fully uses global and multiresolution features to distinguish different scale saliency objects. Huang et al. [11] proposed a semantic-guided SOD model, which locates salient objects by aggregating multiple high-level features and then uses global semantic information to guide the model to obtain saliency detection results. To obtain a semantic context relationship, Zhang et al. [12] proposed an end-to-end dense attention fluid network for saliency detection, which uses the global feature aggregation module to fuse multiscale feature information. Tu et al. [13] proposed a novel SOD model, which uses the high-level attention module to obtain multiscale features and then combines local cues and global information to obtain refined boundary features. To optimize boundary detail information, EMFINet [34] uses the cascaded feature fusion module for multiscale feature aggregation and then uses edge-aware constraints to enhance the feature representation of boundary details. GPNet [35] uses the global perception module to obtain fine-grained global feature information and then uses the attention mechanism and feature distillation structure to reduce the background clutter feature interference. To capture global context information, HFANet [36] uses the hybrid encoder to extract object context information, uses the gated atrous pyramid pooling to capture boundary context features, and introduces the interactive guidance loss for global optimization. To mine the relationship between features, MCCNet [37] constructs the multicontent complementation module to fuse the features of different categories object regions and uses the multivariate loss function to alleviate the problem of category imbalance. RRNet [38] constructs the correlation reasoning module to establish the relationship between spatial and channel dimension features and uses parallel multiscale attention to complete feature complementation.

III. Proposed Method

In Fig. 2, we illustrate the overall framework of our proposed MFENet, which includes GFP, MFE, SFG, and BOM. Different from other SOD methods, MFENet achieves accurate localization and extraction of the object region by fully extracting the multiscale features and deep semantic features contained in the optical RSIs and optimizes the object boundary detection results. For the input RSIs, first, the GFP module is used to obtain the global features and detail features. Second, the MFE module enhances the feature information from the channel and spatial dimensions and suppresses the background feature interference. Third, use the SFG module to enhance the representation ability of MFENet for semantic feature information to improve the detection effect for tiny objects. Finally, the BOM is used to optimize the boundary region detection results from the global dimension. Moreover, we designed a novel cascade loss function that can better optimize the MFENet parameters. The MFENet can achieve MSF and fusion of optical RSIs by means of progressive fusion to obtain accurate SOD results.

A. GFP Module

To fully extract the global feature and detail feature information in RSIs, we propose the GFP module to extract and fuse the global feature in a bottom-up manner. The GFP module is designed as a U-shaped, which can effectively obtain

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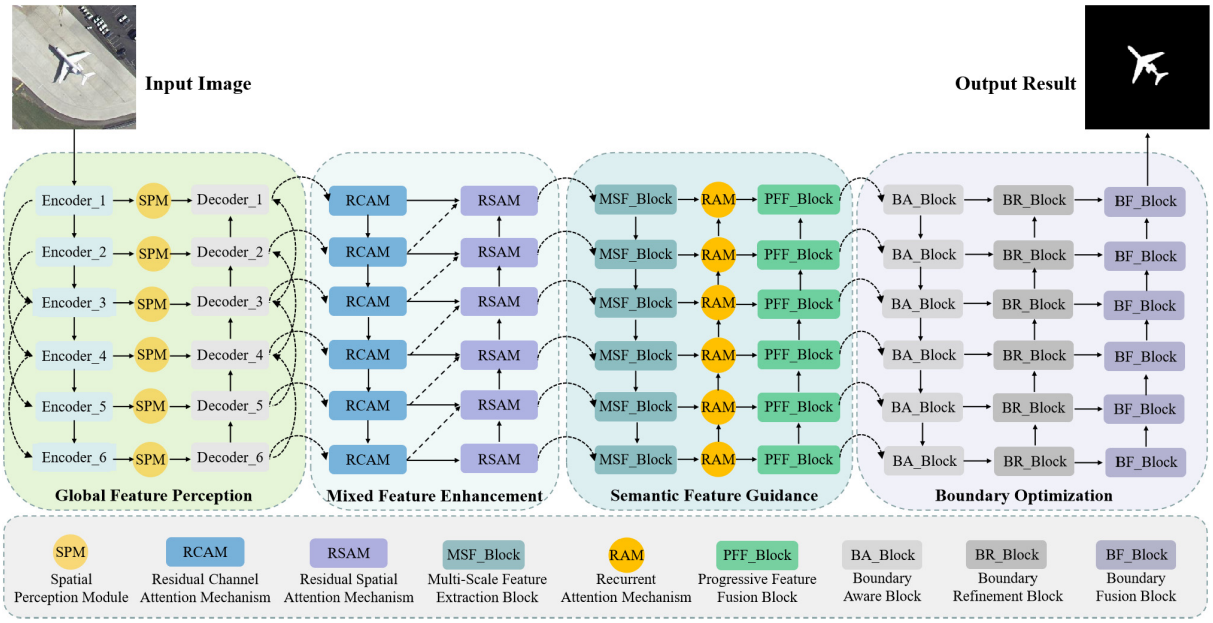


Fig. 2. Overall architecture of the proposed MFENet, including the GFP module, MFE module, FG module, and BOM.

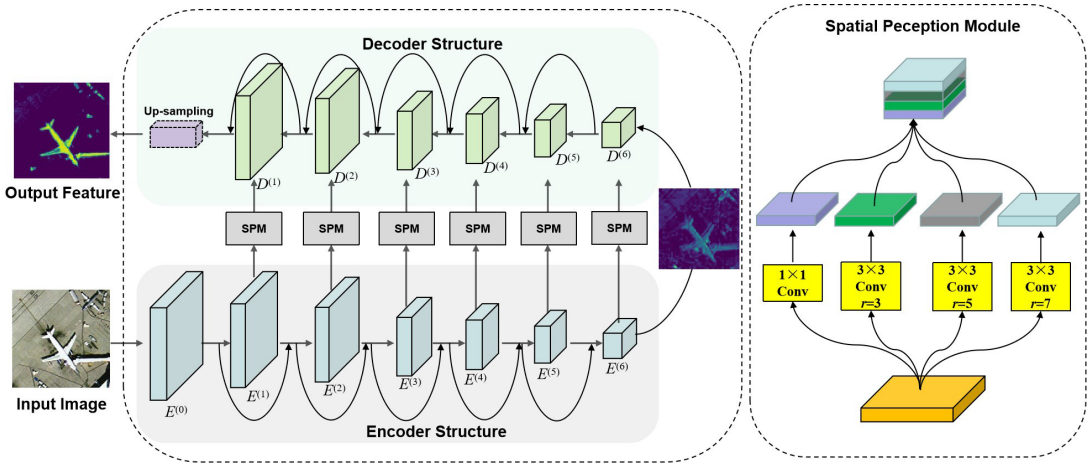


Fig. 3. Structure of GFP module, including the encoder–decoder structure and SPM.

different feature information and fuse different levels' features with a progressive fusion manner. The progressive fusion of different levels' features information can better preserve the details and global features and reduce the feature loss caused by downsampling operations. The structure of the GFP module is shown in Fig. 3, which contains six symmetrical encoder–decoder structures, denoted as $E^{(l)}$ and $D^{(l)}$, where $l \in \{1, 2, 3, 4, 5, 6\}$.

To further obtain the global feature information, we introduced a spatial perception module (SPM) between the encoder and the decoder. The SPM consists of three 3×3 convolutions with different dilated coefficients and one 1×1 ordinary convolution. Then, the feature maps obtained from the convolution layer are spliced, and the batch normalization and ReLU activation function are used to ensure that the output feature map dimension is the same as the input. Specifically, each encoder structure contains different sizes of convolution, pooling, and activation functions; the decoder structure includes deconvolution and upsampling operations. In the encoder structure, the input layer $E^{(1)}$ is constructed using multiple 3×3 convolutions, which can effectively retain the details feature information. The encoder structure $E^{(2)}$–$E^{(4)}$ uses 1×1, 3×3, and 5×5 convolution and max-pooling operations to further extract valuable feature information, respectively, while $E^{(5)}$ and $E^{(6)}$ are composed of residual blocks, which can fully obtain multiscale feature information. The decoder structure consists of six decoder units $D^{(2)}$–$D^{(6)}$, each of which uses the same convolution kernel size as the corresponding encoder unit and performs the upsampling operation to ensure that it can alleviate the feature information loss and restore feature map resolution. The calculation of the encoder–decoder structure is given as follows:

$$D_i^{(l)} = E_o^{(l)} \oplus U^{(l+1)}\left(D_o^{(l+1)}\right), l = 1, 2, 3, 4 \quad (1)$$

$$D_i^{(l)} = E_o^{(l)} \oplus D_o^{(l+1)}, \quad l = 5 \quad (2)$$

$$D_i^{(l)} = E_o^{(l)} \oplus D_o^{SPM}, \quad l = 6 \quad (3)$$

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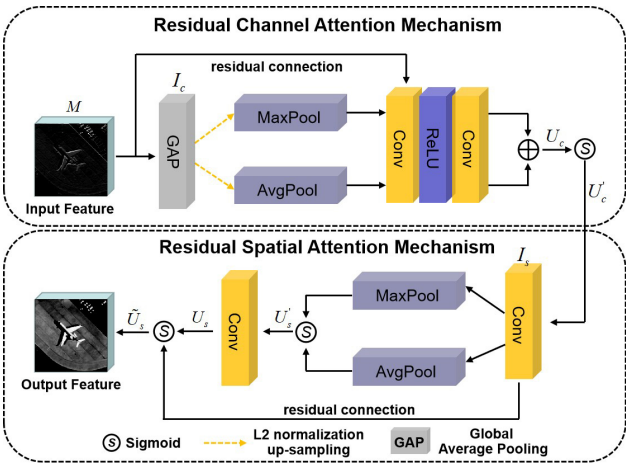


Fig. 4. Structure of MFE module, including the RCAM and RSAM.

where $E_o^{(l)}$ represents the output of the lth encoder unit, $D_i^{(l)}$ and $D_o^{(l)}$ represent the input and output of the lth decoder unit, respectively, $U^{(l+1)}$ represents the $2\times$ upsampling operation on the output of the $(l+1)$th decoder unit, G_o^{SPM} represents the output of the SPM, and $\oplus$ represents the splice operation.

B. MFE Module

To further mine and integrate the detailed and global feature maps output by the GFP module, inspired by squeeze-and-excitation networks (SE-Nets) [39], we propose the MFE module, which effectively integrates the feature by using the residual CAM (RCAM) and the residual spatial attention mechanism (RSAM). The MFE module can enhance the feature information from the channel and spatial dimensions, which effectively suppresses the background feature information interference. As shown in Fig. 4, the MFE series connected the RCAM and RSAM. Specifically, given the input feature maps, MFE can learn the object region feature of the feature maps, obtain the spatial position information, and suppress the background feature information interference.

1) Residual Channel Attention Mechanism: To obtain the relationship between different channels of the input feature map, we use the RCAM to obtain the important feature in the feature matrix. In the convolution operation, restricted by the range of receptive fields, each convolution can only obtain the local feature of the input feature map and cannot obtain the global feature. In the proposed RCAM, inspired by ParseNet [40], the global average pooling (GAP) operation [41] is used to obtain the global feature information of the feature map. The GAP can convert the input 2-D feature map $\boldsymbol{M} \in \mathbb{R}^{H\times W\times C}$ into 1-D feature map $\boldsymbol{I}_c \in \mathbb{R}^{1\times 1\times C}$. To obtain the effective information of different channels in the feature map, two convolution operations are used to extract the nonlinear relationship between each channel, and then, the sigmoid activation function is performed on the feature map output by the convolution operation, which can obtain the logical channel information structure unit $\boldsymbol{U}_c'$. Then, the logical spatial structure unit and the input feature maps are weighted to obtain the channel attention feature matrix $\boldsymbol{U}_c \in \mathbb{R}^{H\times W\times C}$. Finally, to reduce the information loss of the input feature mapping, the channel attention feature mapping and the input feature information are fused by the residual structure to obtain the final channel feature map $\tilde{\boldsymbol{U}}_c \in \mathbb{R}^{H\times W\times C}$

$$\boldsymbol{U}_c = \boldsymbol{U}_c' \times \boldsymbol{M} \tag{4}$$

$$\tilde{\boldsymbol{U}}_c = \boldsymbol{U}_c + \boldsymbol{M} \tag{5}$$

$$\boldsymbol{U}_c' = \sigma(\boldsymbol{W}_2 L(\boldsymbol{W}_1 \boldsymbol{I})) \tag{6}$$

where $L(\cdot)$ represents the Leak ReLU activation function, $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ represent the convolution weight matrix, $\sigma(\cdot)$ represents the sigmoid activation function, and $\boldsymbol{M}$ is the input feature map.

2) Residual Spatial Attention Mechanism : To obtain the spatial relationship between local feature information in the input feature map, we propose the RSAM. Different from the RCAM, the function of RSAM is to focus on the position feature information of the object region in the input feature map. As shown in Fig. 4, to reduce the feature information loss caused by the pooling operation, the 7×7 convolution is first used to reduce the dimensional of the input feature map and obtain the 1-D spatial feature mapping $\boldsymbol{I}_s \in \mathbb{R}^{H\times W\times 1}$. Then, the sigmoid activation function is used to gate control the 1-D feature mapping and obtains the logical spatial information structure unit $\boldsymbol{U}_s'$, and the spatial information structure unit and the 1-D feature mapping are convolved to obtain the spatial attention feature matrix $\boldsymbol{U}_s \in \mathbb{R}^{H\times W\times C}$. Finally, the residual connection is used to fuse the input feature map and the spatial attention feature map, and the spatial attention feature map $\tilde{\boldsymbol{U}}_s \in \mathbb{R}^{H\times W\times C}$ is obtained. The specific calculation process of RSAM is given as follows:

$$\tilde{\boldsymbol{U}}_s = \sigma(\boldsymbol{U}_s + \boldsymbol{I}_s) \tag{7}$$

$$\boldsymbol{U}_s = L\big(\mathrm{Conv}_{1\times 1}\big(\boldsymbol{U}_s'\big)\big) \tag{8}$$

$$\boldsymbol{U}_s'(\boldsymbol{I}_s) = \sigma(\mathrm{Conv}_{7\times 7}(\boldsymbol{I}_s)) \tag{9}$$

where $\sigma(\cdot)$ represents the sigmoid activation function, $L(\cdot)$ represents the LeakReLU activation function, and $\mathrm{Conv}_{7\times 7}(\cdot)$ and $\mathrm{Conv}_{1\times 1}(\cdot)$ represent the 7×7 and 1×1 convolution operations, respectively.

C. SFG Module

In CNNs, semantic features can represent the position information and multiscale feature information of the salient object. Although the GFP module can obtain the global features of salient object regions, it cannot accurately represent the semantic feature information. To fully express the semantic feature information contained in optical RSIs, we propose the SFG module based on MSF block, RAM, and PFF block to guide the model further to obtain semantic features. The MSF block can generate the boundary between the object and background regions, which is beneficial to completing the SOD. The RAM fully uses the advantage of RNN [42] in processing time series data, which can gradually obtain detailed and global features. The PFF block can effectively complete feature fusion with different levels and enhance the model detection performance for tiny objects. The structure of the SFG module is shown in Fig. 5.

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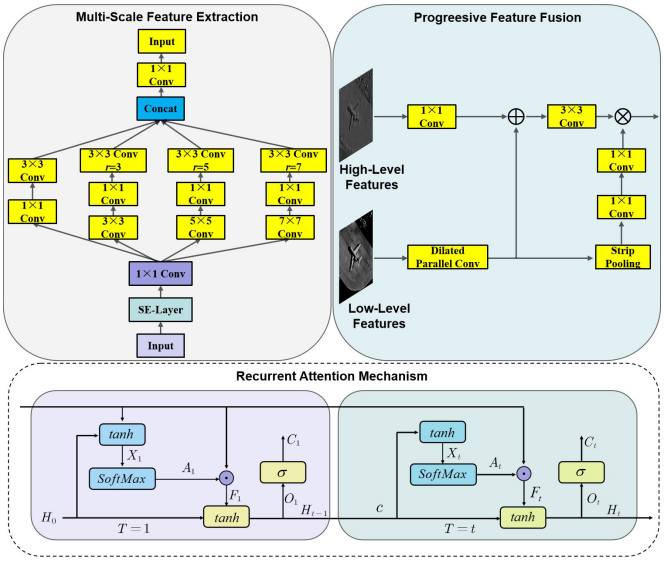


Fig. 5. Structure of SFG module, including the MSF, PFF, and RAM.

1) MSF Block: To solve the salient objects feature representation problem, the proposed MSF block uses multibranch multiscale dilated convolution to extract the deep semantic feature information at different levels, to improve the model feature representation ability for different object categories. Inspired by the SE-Nets [39], we introduce the SE-layer in the MSF block, which can obtain important information of each feature channel by learning manner, therefore achieving effective selection of different features information. The proposed MSF block contains an SE-layer and four convolution branches b_m, where $m = 1, 2, 3$, and 4. Specifically, the SE-layer is first used to select the important channel information of the input feature map; then, the 1×1 convolution is used for feature dimension reduction and four different convolution branches are used for MSF; and finally, the different scale features are fused using the Concat function, and the 1×1 convolution and ReLU activation function are used to output the robust feature information. In the four different scale convolution branches, the first branch b_1 is constructed by 1×1 and 3×3 convolution. The other convolution branches (b_m, $m = 2, 3$, and 4) first use convolution operation with a kernel size of $(2m-1) \times (2m-1)$, and then, the 1×1 and 3×3 convolution with a dilated rate of $(2m-1)$ is used, which can effectively expand the model receptive field range and enhance the feature representation.

2) Recurrent Attention Mechanism: To achieve precise localization and complete contour extraction of salient regions, we propose the RAM, which can effectively extract the location and spatial information of object regions and suppress background feature information interference. The RAM combines RNN and spatial attention mechanism (SAM) [43], which can extract the dependence of input features on time series. The proposed RAM is shown in Fig. 5, which uses the multiscale feature $\boldsymbol{F}_{\text{fuse}}$ output by the MSF block as the input of RAM and then initializes the hidden unit $\boldsymbol{H}_0$, and the refined deep semantic features is obtained by multiple recurrent iterations. In the calculation process of the RAM, it is necessary to input the multiscale feature $\boldsymbol{F}_{fuse}$ into each hidden unit for calculation to realize the multiscale features' enhancement and perform convolution operation between the output results of hidden unit $\boldsymbol{H}_t$ and $\boldsymbol{H}_{t-1}$. The middle layer feature $\boldsymbol{X}_t$ is calculated as follows:

$$\boldsymbol{X}_t = \boldsymbol{V}_a \cdot \tanh(\boldsymbol{W}_a \cdot \boldsymbol{F}_{\text{fuse}} + \boldsymbol{U}_a \times \boldsymbol{H}_{t-1} + b_a) \tag{10}$$

where $\boldsymbol{V}_a$, $\boldsymbol{W}_a$, and $\boldsymbol{U}_a$ represent the weight matrix of convolution kernel and $\boldsymbol{b}_a$ represents the bias term. The Softmax calculation is performed on each pixel of the feature map $\boldsymbol{X}_t$ to obtain the normalized spatial attention feature map A_t^{xy}

$$A_t^{xy} = \text{Soft}\max\left(X_t^{xy}\right) = \frac{\exp\left(X_t^{xy}\right)}{\sum_x \sum_y \exp\left(X_t^{xy}\right)} \tag{11}$$

where A_t^{xy} is the attention value on position (x, y). The attention feature map $\boldsymbol{A}_t$ and the input feature $\boldsymbol{F}_{\text{fuse}}$ are dot multiplied on each channel, and the attention feature map $\boldsymbol{F}_t$ is obtained. The calculation process is given as follows:

$$\boldsymbol{F}_t = \boldsymbol{A}_t \odot \boldsymbol{F}_{\text{fuse}}. \tag{12}$$

The attention feature map $\boldsymbol{F}_t$ is used as the input of RNN and combined with $\boldsymbol{H}_{t-1}$ to update the hidden unit $\boldsymbol{H}_t$

$$\boldsymbol{H}_t = \tanh(\boldsymbol{W}_h \times \boldsymbol{F}_t + \boldsymbol{U}_h \times \boldsymbol{H}_{t-1}) \tag{13}$$

where $\boldsymbol{W}_h$ and $\boldsymbol{U}_h$ represent the convolution kernel weight matrix. The updated hidden unit $\boldsymbol{H}_t$ is used to update the output unit $\boldsymbol{O}_t$ and memory unit $\boldsymbol{C}_t$

$$\boldsymbol{O}_t = \boldsymbol{V}_o \times \boldsymbol{H}_t + \boldsymbol{b}_o \tag{14}$$

$$\boldsymbol{C}_t = \text{sigmoid}(\boldsymbol{O}_t) \tag{15}$$

where $\boldsymbol{V}_o$ is the convolution kernel weight and $\boldsymbol{b}_o$ represents the bias term.

3) PFF Block: The effective fusion of high- and low-level features is conducive to SOD. The common feature fusion methods use feature splicing to fuse different levels' features, resulting in detailed feature information loss. To effectively fuse features with different scales, we propose the PFF block for fusing high- and low-level feature information. For the high-level features, first, the dilated convolution [44] with different dilated coefficients is used for deep feature extraction, and the extracted features are classified according to the corresponding channels; then, the strip pooling [45] operation is used to obtain the spatial dependence between different feature channels, and 1×1 convolution is used for feature normalization. For the low-level features, the 1×1 convolution is first used to normalize the features, and it is weighted with the high-level features extracted by dilated convolution; then, the 3×3 convolution is used for feature enhancement to obtain the multiscale deep semantic features; and finally, the low-level feature is fused with the high-level feature using the corresponding element multiplication operation

$$\boldsymbol{F}_1 = \text{Conv}_{1\times1}(f_{low}) + P_\text{Conv}\left(f_{\text{high}}\right) \tag{16}$$

$$\boldsymbol{F}_2 = \text{Conv}_{1\times1}\left(SP\left(P_\text{Conv}\left(f_{\text{high}}\right)\right)\right) \tag{17}$$

$$\boldsymbol{F}_{\text{fuse}} = \text{Conv}_{3\times3}(\boldsymbol{F}_1) \otimes \text{Conv}_{1\times1}(\boldsymbol{F}_2) \tag{18}$$

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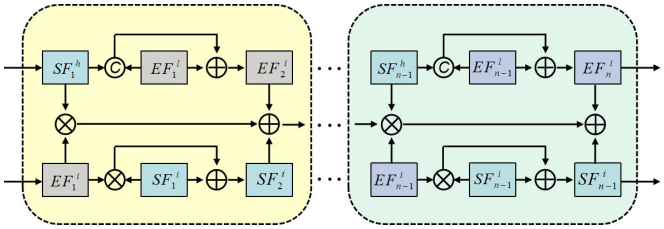


Fig. 6. Structure of boundary-aware block.

where $\boldsymbol{F}_1$ is the low-level feature fusion process, $\boldsymbol{F}_2$ is the high-level feature fusion process, $P_\mathrm{Conv}(\cdot)$ represents the parallel dilated convolution, $SP(\cdot)$ represents the strip pooling operation, and $\otimes$ is the corresponding element multiplication.

D. Boundary Optimization Module

The convolution and pooling operations in the CNNs module will cause the boundary features loss, reducing the model detection accuracy for the object boundary To alleviate the boundary feature loss, we propose the BOM module, which contains a boundary-aware block, boundary refinement block, and boundary fusion block.

1) Boundary-Aware Block: The refined extraction of boundary features plays an important role in improving the accuracy of remote sensing SOD. To obtain the boundary feature information of the target region, the stacked cross refinement network [46] uses the stacked cross refinement unit (CRU) to extract the object boundary features and salient features effectively. The semantic information is beneficial for locating salient objects, and texture features can refine the boundary of salient object regions. Based on the CRU structure, we propose a boundary-aware block that uses high-level semantic information and low-level texture information. The structure of boundary-aware block is shown in Fig. 6, which is designed as an asymmetrical cross structure that facility cascade and multiplication operations of different channel features to realize the fusion of high-low level feature information. In the boundary-aware block, the boundary features and saliency maps can supervise each other and progressively refine the extracted feature information in a stacked manner

$$\mathrm{SF}_n^i = \mathrm{SF}_{n-1}^i + f\left(\mathrm{SF}_{n-1}^i, \mathrm{EF}_{n-1}^l\right) \tag{19}$$

$$\mathrm{EF}_n^i = \mathrm{EF}_{n-1}^i + g\left(\mathrm{EF}_{n-1}^i, \mathrm{SF}_{n-1}^h\right) \tag{20}$$

$$f = \mathrm{Conv}\left(\mathrm{Cat}\left(\mathrm{SF}_{n-1}^i, \mathrm{EF}_{n-1}^l\right)\right) \tag{21}$$

$$g = \mathrm{Conv}\left(EF_{n-1}^i \otimes SF_{n-1}^h\right) \tag{22}$$

where SF_n^i and EF_n^i represent the salient features and boundary features in the nth boundary-aware block after i layer feature extraction, respectively; EF_{n-1}^l and SF_{n-1}^h represent the boundary features and salient features of low- and high-level feature maps in the $(n-1)$th boundary-aware block, respectively; and f and g represent the correlation mutual supervision function of salient features and boundary features, respectively.

2) Boundary Refinement Block: To further obtain the complete boundary feature information, we propose a boundary refinement block for boundary feature refinement and enhancement. Specifically, as shown in Fig. 7, the boundary refinement block uses 3×3 convolution for feature extraction, and the ReLU activation function is used to enhance the nonlinearity of the block; then, the hybrid dilation convolution (HDC) composed of different receptive field dilated convolution is used for multiscale boundary feature extraction, and the *Concat* function and 1×1 convolution are used for multiscale feature fusion and scale normalization; and finally, the residual connection and corresponding element addition operation are used to stitching fusion the input features. In the boundary refinement block, different from the limited receptive field range of traditional convolution, the hybrid dilated convolution can expand the receptive field range without increasing the calculation parameter to obtain more boundary refinement features. Moreover, to make the receptive field range of the hybrid dilated convolution cover all the pixels of the feature map, we use the 3×3 convolution with dilated coefficients $r \in \{1, 2, 5\}$ to construct the HDC, and the 1×1 convolution is used for feature fusion. The calculation process of the boundary refinement module is given as follows:

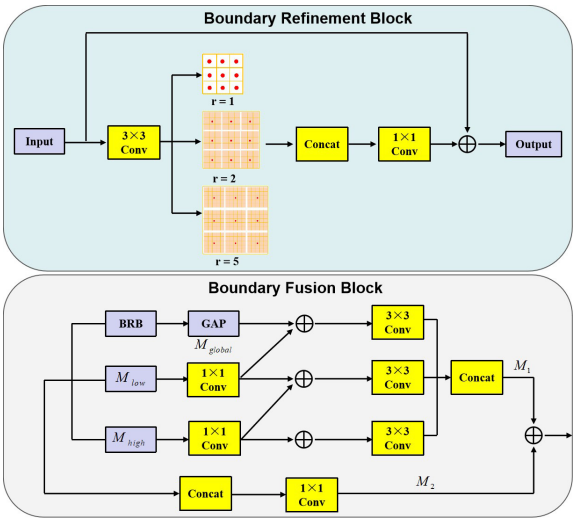


Fig. 7. Structure of boundary refinement and fusion blocks.

$$X_1 = \mathrm{Conv}_{3\times3}(X_I);\ X_2 = \mathrm{Conv}_{3\times3,r=1}(X_1) \tag{23}$$

$$X_3 = \mathrm{Conv}_{3\times3,r=2}(X_1);\ X_4 = \mathrm{Conv}_{3\times3,r=5}(X_1) \tag{24}$$

$$X_5 = \mathrm{Conv}_{1\times1}(\mathrm{Cat}(X_2, X_3, X_4)) \tag{25}$$

$$X_O = X_5 \oplus X_I \tag{26}$$

where X_I represents the input feature map, X_O represents the output boundary refinement feature, $\mathrm{Conv}_{3\times3}(\cdot)$ is the 3×3 convolution operation, and $\mathrm{Conv}_{3\times3,r=5}(\cdot)$ represents 3×3 dilation convolution with dilated coefficient r.

3) Boundary Fusion Block: Boundary features of different levels contain different feature information distribution, the low-level features contain spatial structure information with high resolution, whereas high-level features contain semantic information with low resolution. To use the global boundary feature to guide the model for saliency inference, we propose a boundary feature fusion block that highlights the boundary features and suppresses background interference. The structure

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of boundary fusion block is shown in Fig. 7, which includes three stages: multiscale boundary feature fusion, low-level feature fusion, and high-level feature fusion. For multiscale boundary feature fusion, first, the GAP is performed on the feature map output by the boundary refinement block to obtain the global context boundary feature and then calibrates the feature map channel information to generate a mask with global information, and finally, the convolution operation is used to fuse with the high-to-low-level feature information. For low- and high-level features, the convolution is used to generate the corresponding mask; then, the masks of different levels are multiplied for feature complementation, and feature cascading is performed using the fusion operation; and finally, the convolution operation is used to obtain the fusion feature map M_1. In addition, to obtain more abundant global boundary features, the global feature map M_2 is obtained by cascade and convolution operations, and the boundary feature fusion result is obtained by adding M_1 and M_2

$$M_g = \text{Conv}(M_{low}) \times \text{Mask}\left(M_{\text{global}}\right) \tag{27}$$

$$M_l = \text{Conv}\left(M_{\text{high}}\right) \times \text{Mask}(M_{\text{low}}) \tag{28}$$

$$M_h = \text{Mask}\left(M_{\text{high}}\right) \times \text{Conv}(M_{\text{low}}) \tag{29}$$

$$M_a = C\left(M_g, M_l, M_h\right) \tag{30}$$

$$M_b = C\left(\left(\text{Conv}\left(M_{\text{global}}, M_{\text{low}}, M_{\text{high}}\right)\right)\right) \tag{31}$$

$$M_w = \text{Add}(\text{Conv}(M_a, M_b)) \tag{32}$$

where M_{low}, M_{high}, and M_{global} represent low-level boundary features, high-level boundary features, and global boundary features, respectively. M_l is the fusion result of low-level feature masks with high-level feature, M_h is the fusion result of high-level feature masks with low-level feature, Conv(·) represents the convolution operation, Mask(·) represents the corresponding mask generated by different level features, and Add(·) is the corresponding elements' addition operation.

E. Cascade Loss Function

Since the MFENet consists of saliency object location, saliency region extraction, and boundary extraction, inspired by the previous work in [27], we propose the cascade loss function for the model train. The cascade loss function includes classification loss, structural similarity (SSIM) loss, and boundary optimization loss, which are defined as

$$L_{\text{cascade}} = L_{\text{cls}} + L_{\text{ssim}} + L_{\text{boundary}} \tag{33}$$

where L_{cls} represents the classification loss, L_{ssim} represents the SSIM loss, and L_{boundary} represents the boundary optimization loss.

1) Classification Loss: In the process of locating the salient object region, since the number of negative samples is higher than that of positive samples, we use the classification loss function proposed in S4Net [47] to solve the sample imbalance in the model training process by calculating the positive samples loss and negative samples loss

$$L_{\text{cls}} = -\left(\frac{1}{N_p}\sum_{i\in P}\ln p_i + \frac{1}{N_N}\sum_{j\in N}\ln\left(1-p_j\right)\right) \tag{34}$$

where N_P and N_N represent the number of positive samples and negative samples, respectively, and P_i represents the positive and negative samples probability of the ith input image.

2) Structural Similarity Loss: The SSIM loss [48] is a loss function proposed for image quality assessment, which can capture the structural details information of the input image, so it is introduced into the cascade loss function to learn the structural information of the saliency object area. Define $x = \{x_j : j = 1, \ldots, N\}$ and $y = \{y_j : j = 1, \ldots, N\}$ as detection result and corresponding ground truth, and the SIMM loss is calculated as follows:

$$L_{\text{ssim}} = 1 - \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{35}$$

where μ and σ represent the standard deviation and variance of x and y, respectively, σ_{xy} represents the variance, and C_1 and C_2 are the custom parameters that prevent the denominator being zero.

3) Boundary Optimization Loss: Since the function of boundary optimization block is to better guide the model for boundary feature extraction, thereby improving the SOD accuracy, the mean square error (mse) [49] is introduced as the boundary optimization loss function

$$L_{\text{boundary}} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - f(x_i)\right)^2 \tag{36}$$

where for the ith input image, $f(x_i)$ represents the saliency detection results predicted by the model, y_i represents the corresponding ground truth, and N is the number of input images.

IV. Experiments

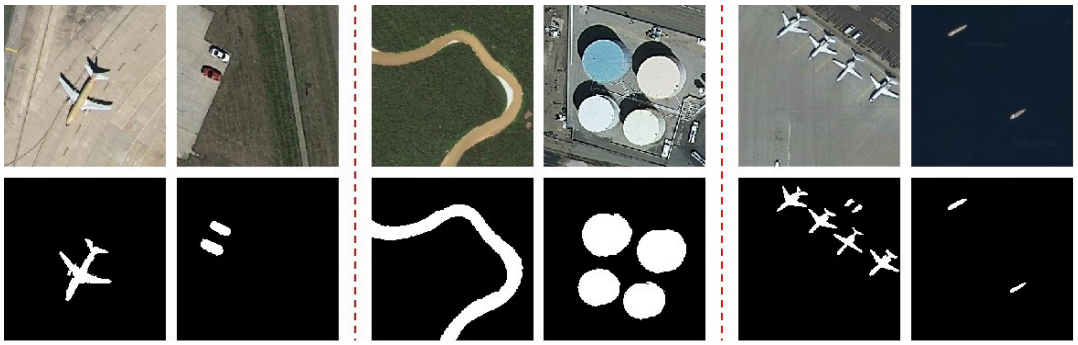
In this section, we introduce three different optical RSIs' benchmark dataset (ORSSD [9], EORSSD [12], and ORSI-4199 [13]) for SOD and then illustrate the algorithm performance evaluation metrics, training strategies, and implementation details. Then, the proposed MFENet is compared with other state-of-the-art methods to demonstrate its advantage, and ablation studies are carried out to verify the rationality of the model structure. Finally, to illustrate the defects of MFENet, we conducted a failure case analysis.

A. Benchmark Dataset

In the experiment, we use ORSSD, EORSSD, and ORSI-4199 optical RSIs datasets as benchmarks. The details of different datasets are introduced as follows.

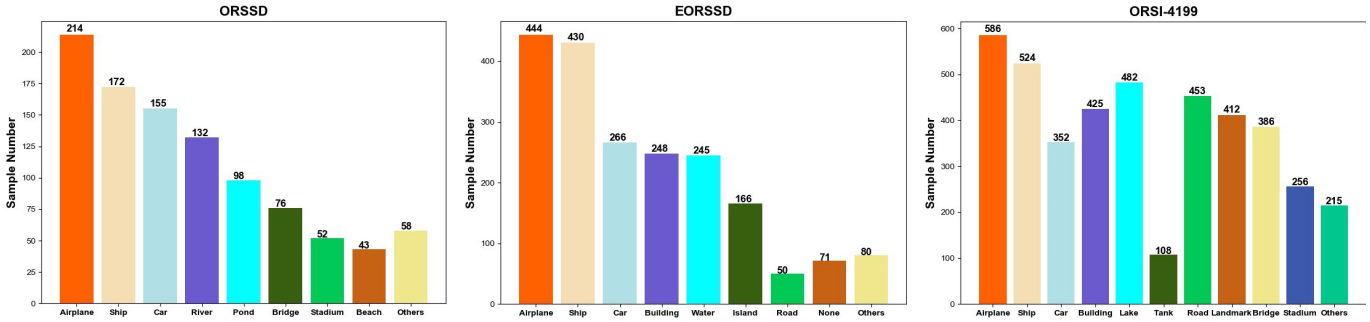
1) ORSSD: The ORSSD dataset is the first dataset for optical RSIs' SOD, containing 800 original optical RSIs and corresponding manual annotations. Since ORSSD contains multiclass targets (airplane, ship, car, river, and so on) with different spatial resolutions (1264 × 987, 800 × 600, and 256 × 256), and the background regions contain many interference objects (shadow, trees, and buildings), which poses great challenges for accurate detection salient objects. In addition, there are large size variations between different objects, such as buildings and cars. Fig. 8(a) shows the

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(a) (b) (c)

Fig. 8. Visualization results of different benchmark datasets. (a) ORSSD. (b) EORSSD. (c) ORSI-4199.



(a) (b) (c)

Fig. 9. Number statistical information of different benchmark datasets. (a) ORSSD dataset. (b) EORSSD. (c) ORSI-4199.

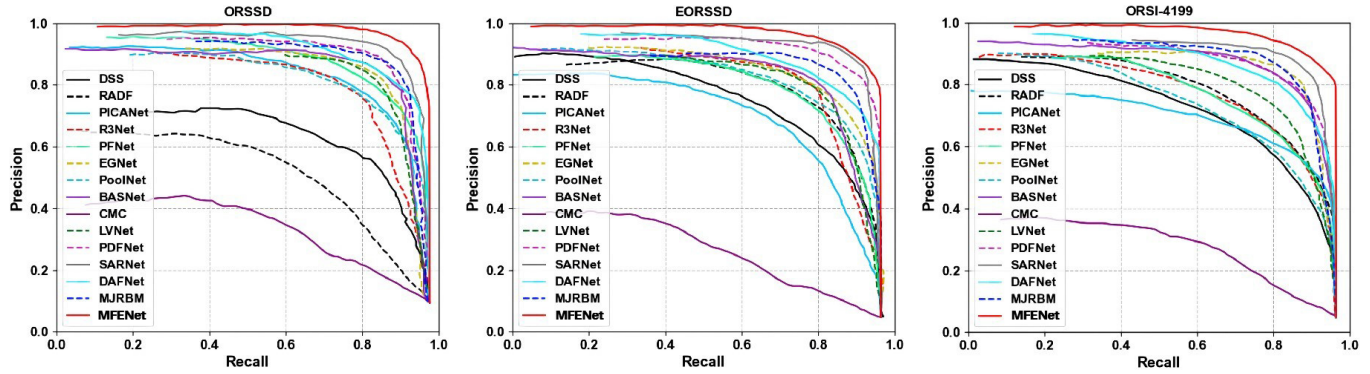


Fig. 10. P–R curves of different methods on the testing dataset of the ORSSD, EORSSD, and ORSI-4199.

sample images and corresponding annotations of ORSSD, and the statistical information of different objects is shown in Fig. 9(a).

2) EORSSD: The EORSSD dataset is an extension of the ORSSD dataset. The EORSSD contains 2000 optical RSIs images and corresponding pixelwise annotation information. Fig. 9(b) shows the statistical information of EORSSD. Compared with ORSSD, the EORSSD contains many tiny target categories (car, ship, aircraft, and so on) and complex scenarios (road, water, and island). As shown in Fig. 8(b), the single image contains multiple categories of salient objects. Similarly, EORSSD has the problem of large sample size variation and the same object category with different sizes. Furthermore, EORSSD contains many imaging interferences, such as clouds and illumination, which destroy image integrity and pose challenges to SOD.

3) ORSI-4199: ORSI-4199 is a large-scale dataset for validating the performance of optical RSIs SOD algorithm, which contains 4199 RSIs in different scenarios. The ORSI-4199 dataset has the characteristics of multiple object categories (car, bridge, road, lake, ship, and so on), object size small (the ratio between the object and the entire image is not larger than 0.3), strong background interference, complex texture structure, and incomplete salient object. Fig. 8(c) shows the original images and manual annotation of the ORSI-4199 dataset, it can be seen that the image scene is more complex, and the object size varies. Fig. 9(c) shows the statistical information of different object categories.

B. Evaluation Metrics

To verify the performance of the proposed MFENet, we employ a precision-recall (P–R) curve, F-measure, mean

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absolute error (MAE) score, and S-measure as evaluation metrics. For the P–R curve, as it consists of precision and recall, we first threshold the generated saliency maps into the binary saliency maps by using a series of fixed integers from 0 to 255 and then compared the binary mask with the ground truth to obtain precision and recall. In the P–R curve, the closer the P–R curve is to the coordinate points of (1, 1), the better the performance achieves.

1) F-Measure: The F-measure is a comprehensive measurement index used to measure the model performance, which is defined as the weighted harmonic mean of precision and recall [50]. The calculation of F-measure is given as follows:

$$F_\beta = \frac{\left(1+\beta^2\right)\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{37}$$

where β^2 is set to 0.3 to emphasize the precision over recall and the larger F-measure indicates the better performance.

2) MAE: It is to measure the mean absolute difference between the predicted saliency map and the corresponding ground truth [51], which is represented as

$$\text{MAE} = \frac{1}{w \times h} \sum_{i=1}^{w} \sum_{j=1}^{h} |S(i, j) - G(i, j)| \tag{38}$$

where w and h represent the width and height of the image, respectively, and the small MAE indicates the better performance.

3) S-Measure: The S-measure is used to measure the SSIM between the predicted saliency map and ground truth [52]. The S-measure is represented as

$$S_m = \alpha \times S_o + (1-\alpha) \times S_r \tag{39}$$

where α is set to 0.5 for assigning equal contribution to object perception S_o and region perception S_r, and the large S-measure indicates the better performance.

C. Training Strategies and Implementation Details

The proposed MFENet is built using the Pytorch framework on a PC with Ubuntu 18.04 operating system, and the NVIDIA GeForce RTX 3090 GPU with 24-GB memory is used for acceleration. In the training phase, the Xavier strategy is used to initialize the weight parameters, and the stochastic gradient descent (SGD) is used to optimize the weight. The batch size is set to 24, and the epoch is set to 100. In the first 40 epochs, the learning rate is fixed to $1e^{-4}$, and then, the learning rate is automatically adjusted by an exponential decay. The number of samples in the training set and testing set for three benchmark datasets is different. For the ORSSD dataset, we use 600 images as the training set and 200 images as the testing set. For the EORSSD dataset, we adopt 1400 images as the training set and 600 images as the testing set. For the ORSI-4199 dataset, 2000 images are used as the training set and 2199 images as the testing set.

D. Comparison With State-of-the-Art Methods

To verify the effectiveness and performance of the proposed method, we compare the proposed MFENet with 14 existing SOD methods on the testing dataset of ORSSD, EORSSD, and ORSI-4199. The compared methods include SOD for optical RSIs (CMC [33], LVNet [9], PDFNet [10], SARNet [11], DAFNet [12], and MJRBM [13]) and SOD for NSIs (DSS [20], RADF [21], PICANet [22], R3Net [23], PFAN [24], EGNet [25], PooLNet [26], and BASNet [27]). All the results are generated by the source codes or provided by the authors. Moreover, because the compared methods are all deep learning-based, the same training strategy as the proposed method is used to ensure fairness.

1) Quantitative Comparison: Table I and Fig. 10 show the quantitative comparison results of the proposed MFENet and other state-of-the-art methods on three different benchmark datasets. It can be seen from Table I and Fig. 10 that our MFENet outperforms the existing methods in terms of the evaluation metrics P–R curve, F-measure, MAE score, and S-measure. Specifically, compared with the MJRBM [13], the proposed MFENet exceeds it by 7.31% and 1.80% on the ORSSD dataset with the F-measure and S-measure, respectively. The reason is that MFENet can fully extract the global context features of optical RSIs, and the BOM can ensure the structural integrity of the extracted object region. Compared with the DAFNet [12], our MFENet exceeds it by 1.81% and 2.77% on the evaluation metrics of F-measure and S-measure, respectively. Although DAFNet uses the encoder–decoder structure to alleviate the feature information loss, it cannot solve the boundary feature loss problem. In addition, compared with other optical RSIs SOD methods (CMC [33], LVNet [9], PDFNet [10], and SARNet [11]), the proposed MFENet is significantly better than them on F-measure and S-measure ([50.99%, 33.67%], [10.02%, 6.50%], [5.56%, 3.53%], and [5.46%, 1.15%], respectively).

For the EORSSD dataset, Table I shows that our MFENet still achieved better results on evaluation metrics F-measure and S-measure. Compared with the MJRBM [13] and DAFNet [12] method, the F-measure increased 3.30% and 2.98%, and the S-measure increased 1.34% and 0.62%, respectively. In addition, for the NSIs' SOD methods, since the feature information contained in RSIs cannot be fully extracted, it has not achieved competitive results.

For the ORSI-4199 dataset, although it contains many complex scenarios, our MFENet still gets the best results, indicating that the proposed method can be robust against the interference of scene changes. Compared with the MJRBM [13], the evaluation metrics, F-measure, MAE score, and S-measure, increased 5.84%, 0.15%, and 0.79%, respectively. Although the MJRBM uses the boundary optimization strategy, its feature aggregation process causes valuable information loss, while MFENet can retain feature information using the proposed SFG module. Compared with the DAFNet [12], the evaluation metrics, F-measure, MAE score, and S-measure, increased 5.42%, 1.29%, and 2.44%, respectively. As for other RSIs' SOD methods, they have not achieved competitive results due to the great influence of complex scenes.

The P–R curve, precision, and recall of different methods on three benchmark datasets are shown in Figs. 10 and 11. The results in Fig. 10 show that the P–R curves of our method

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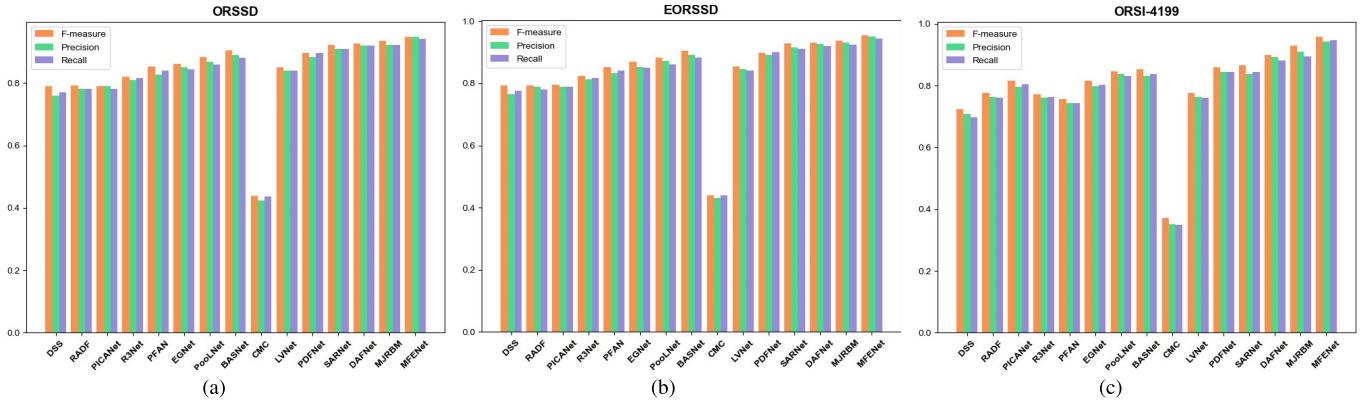


Fig. 11. Comparison of F-measure, precision, and recall among different methods on (a) ORSSD, (b) EORSSD, and (c) ORSI-4199.

TABLE I
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT METHODS ON THE TESTING DATASET OF ORSSD, EORSSD, AND ORSI-4199

	ORSSD Dataset			EORSSD Dataset			ORSI-4199 Dataset		
	$F_\beta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_m \uparrow$
DSS [20]	0.7793	0.0357	0.8371	0.7254	0.0178	0.7926	0.7051	0.0154	0.7738
RADF [21]	0.7845	0.0364	0.8152	0.7821	0.0168	0.8175	0.7536	0.0124	0.7683
PICANet [22]	0.7843	0.0342	0.7954	0.7769	0.0296	0.8041	0.7954	0.0286	0.8157
R3Net [23]	0.8124	0.0382	0.8247	0.7783	0.0172	0.8047	0.7526	0.0137	0.7835
PFAN [24]	0.8415	0.0552	0.8574	0.7784	0.0162	0.8471	0.7357	0.0128	0.7683
EGNet [25]	0.8537	0.0224	0.8615	0.8134	0.0173	0.8642	0.7938	0.0216	0.8135
PooLNet [26]	0.8745	0.0247	0.8512	0.8437	0.0324	0.8574	0.8243	0.0275	0.8324
BASNet [27]	0.8927	0.0279	0.8846	0.8564	0.0175	0.8654	0.8276	0.0298	0.8343
CMC [33]	0.4317	0.1265	0.6038	0.3762	0.1047	0.5800	0.3576	0.1357	0.6215
LVNet [9]	0.8414	0.0207	0.8815	0.8051	0.0145	0.8645	0.7542	0.0301	0.8431
PDFNet [10]	0.8860	0.0149	0.9112	0.8557	0.0174	0.8872	0.8347	0.0246	0.8274
SARNet [11]	0.8870	0.0100	0.9350	0.8570	0.0080	0.9290	0.8375	0.0275	0.8173
DAFNet [12]	0.9235	0.0106	0.9188	0.9060	0.0053	0.9185	0.8743	0.0213	0.8972
MJRBM [13]	0.8685	0.0372	0.9285	0.9025	0.0145	0.9113	0.8701	0.0099	0.9137
MFENet	0.9416	0.0072	0.9465	0.9358	0.0065	0.9247	0.9285	0.0084	0.9216

on three benchmark datasets are closer to the coordinate (1, 1), indicating that MFENet has a better performance than other methods. It can be seen from Fig. 11 that the proposed MFENet has high precision and recall on three datasets and small differences between the two evaluation metrics, indicating that our method achieved better results on the RSIs SOD task. The reason is that MFENet can obtain sufficient multiscale features and global semantic features by the proposed GFP module, MFE module, and SFG module, which can effectively guide the model complete accurate SOD.

2) Qualitative Comparison: The qualitative comparison results of different methods are shown in Fig. 12. To comprehensively analyze the performance advantages of our method, we select different challenging scenarios for visual comparative analysis, including small-scale salient objects (first, four, seven, and nine rows), narrow salient objects (two and six rows), low-contrast salient objects (three and eight rows), and big-scale salient objects (five and ten rows). Compared to other different methods, the proposed MFENet exhibits advantages and competitiveness in the following aspects.

a) Advantages in location accuracy: As shown in Fig. 12, in the first, four, seven, and nine images, our method can achieve precise localization of small salient objects (SSOs) and suppress background interference. Compared with other methods, such as MJRBM [13] and DAFNet [12], although these methods can locate the object region, they have poor detection accuracy of the edge region and are seriously interfered with by background information. Compared with SARNet, PDFNet, and CMC, these methods cannot achieve precise localization of the object region, such as the first-row image, which shows detection errors for car objects, indicating that these methods cannot complete salient detection of small-size remote sensing objects. In terms of NSIs SOD methods, due to the severe interference of background information, they have problems, such as positioning deviation and poor salient detection effect of boundary region.

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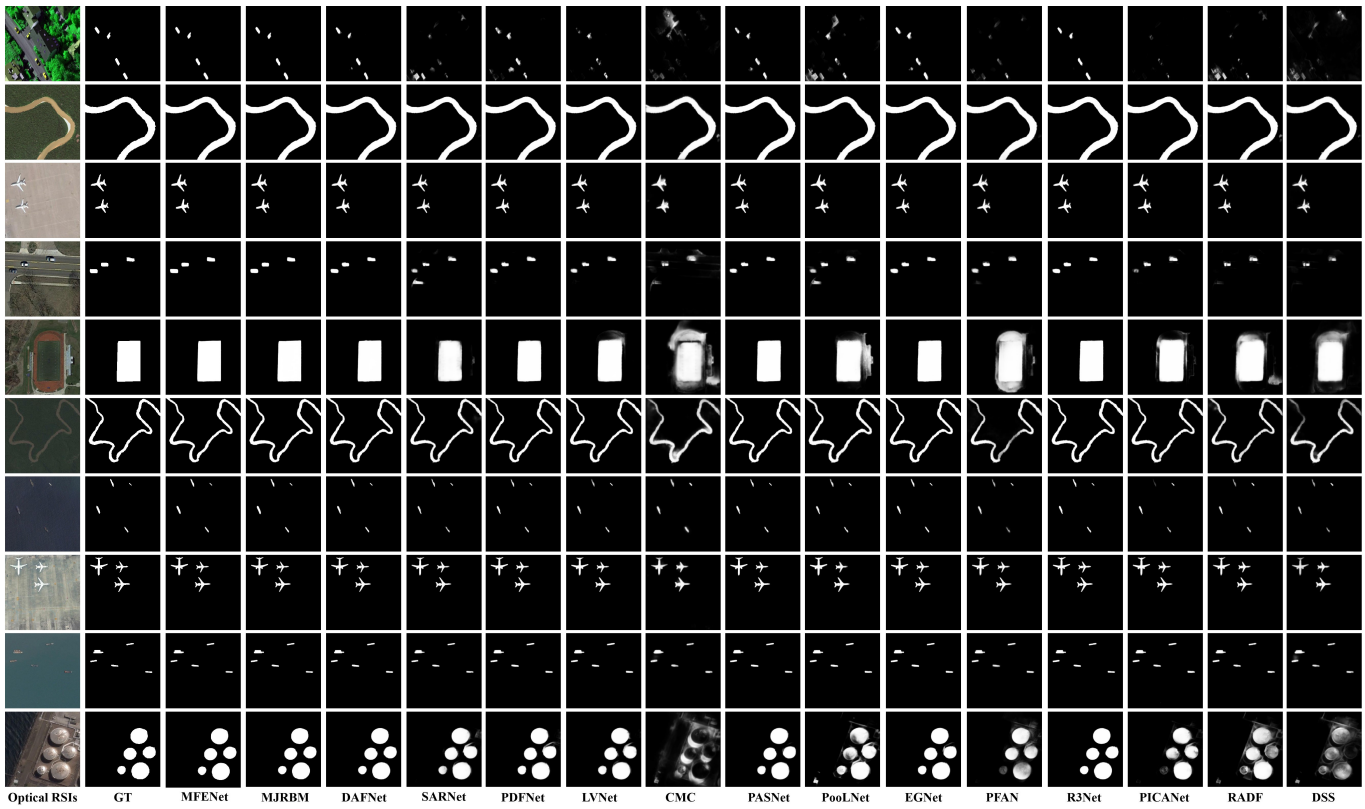


Fig. 12. Visual comparison results of different methods on ORSSD, EORSSD, and ORSI-4199.

TABLE II

F-MEASURE EVALUATION BASED ON SCENE ATTRIBUTE ON THE ORSI-4199 DATASET. BSO: BIG SALIENT OBJECT. SSO: SMALL SALIENT OBJECT. OC: OFF CENTER. CSO: COMPLEX SALIENT OBJECT. CS: COMPLEX SCENE. NSO: NARROW SALIENT OBJECT. MSO: MULTIPLE SALIENT OBJECTS. LCS: LOW CONTRAST SCENE. ISO: INCOMPLETE SALIENT OBJECT. TOP THREE RESULTS ARE MARKED IN GREEN, BLUE, AND RED

Scene	CMC [33]	LVNet [9]	PDFNet [10]	SARNet [11]	DAFNet [12]	MJRBM [13]	MFENet
BOS	0.3941	0.8136	0.8862	0.8743	0.9014	0.9186	0.9278
SSO	0.2863	0.7124	0.7864	0.7912	0.8245	0.8232	0.8452
OC	0.2914	0.7236	0.7931	0.7743	0.8042	0.8457	0.8549
CSO	0.3762	0.7864	0.8546	0.8415	0.8756	0.8913	0.9124
CS	0.3862	0.7932	0.8628	0.8726	0.8913	0.8892	0.9072
NSO	0.3254	0.7415	0.8417	0.8536	0.8675	0.8785	0.8915
MSO	0.3415	0.7683	0.8243	0.8371	0.8714	0.8624	0.8721
LCS	0.3152	0.7385	0.7956	0.7815	0.8136	0.8047	0.8175
ISO	0.3825	0.7792	0.8673	0.8724	0.8872	0.9075	0.9193
Avg	0.3443	0.7619	0.8345	0.8331	0.8597	0.8690	0.8831

b) Advantages in content integrity: In the sixth image, due to the river containing complex structural information and relatively narrow tributaries, some methods, such as SARNet, PDFNet, LVNet, and CMC, fail to precision detection. In the image of five and ten rows, which contains irregular salient object regions, our method can accurately detect the object details. In contrast, compared methods, such as SARNet [11], PDFNet [10], LVNet [9], and CMC [33], suffer from edge details information loss and serious interference by background information. For NSIs SOD methods such as PooLNet [26], PFAN [24], PICANet [22], RADF [21], and DSS [20], due to their poor ability to extract multiscale features contained in RSIs, these methods cannot achieve accurate detection of irregular object regions. Experimental results show that our method can retain the complete object structure detail information and achieve accurate SOD.

c) Advantages in challenging scenarios: For challenging complex scenarios, such as the image of three, eight, and nine rows, the image contains multiple objects, especially the third and eighth, which are affected by low contrast. As shown in the detection results of Fig. 12, although the complex scene impacts different detection methods, our method still

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TABLE III
F-MEASURE EVALUATION BASED ON CATEGORY ATTRIBUTE ON THE ORSSD DATASET

Category	CMC [33]	LVNet [9]	PDFNet [10]	SARNet [11]	DAFNet [12]	MJRBM [13]	MFENet
Airplane	0.3572	0.8145	0.8541	0.8756	0.8975	0.9116	0.9538
Ship	0.2915	0.8076	0.8216	0.8917	0.9114	0.9035	0.9357
Car	0.3124	0.7452	0.7985	0.8614	0.9012	0.9047	0.9265
River	0.3715	0.8369	0.8573	0.8537	0.8935	0.9124	0.9358
Pond	0.3684	0.8953	0.8614	0.8841	0.9015	0.9238	0.9563
Bridge	0.3579	0.8231	0.8326	0.8875	0.8983	0.9073	0.9478
Stadium	0.3852	0.8415	0.8572	0.8916	0.9143	0.9136	0.9386
Beach	0.3672	0.8342	0.8614	0.8673	0.8928	0.9124	0.9425
Avg	0.3514	0.8247	0.8430	0.8766	0.9013	0.9112	0.9421

TABLE IV
F-MEASURE EVALUATION BASED ON QUANTITY ATTRIBUTE ON THE EORSSD DATASET

Quantity	CMC [33]	LVNet [9]	PDFNet [10]	SARNet [11]	DAFNet [12]	MJRBM [13]	MFENet
1	0.3726	0.8236	0.8673	0.8972	0.9064	0.9125	0.9238
2	0.3524	0.7862	0.8514	0.8843	0.8936	0.8758	0.9351
3	0.3126	0.8173	0.8321	0.8146	0.8745	0.8863	0.9257
4	0.3108	0.8326	0.8426	0.8451	0.8652	0.9042	0.9186
$\geqslant 5$	0.2879	0.8217	0.8254	0.8369	0.8478	0.8845	0.9124

TABLE V
F-MEASURE EVALUATION BASED ON SIZE ATTRIBUTE

Quantity	CMC [33]	LVNet [9]	PDFNet [10]	SARNet [11]	DAFNet [12]	MJRBM [13]	MFENet
$\leqslant 1\%$	0.2275	0.7731	0.8514	0.8615	0.8752	0.8914	0.9385
1∼10%	0.2864	0.8157	0.8747	0.8848	0.9125	0.9032	0.9452
10∼20%	0.3125	0.8732	0.8975	0.8932	0.9413	0.9381	0.9578
20∼30%	0.3247	0.8716	0.8735	0.9175	0.9426	0.9523	0.9624
30∼40%	0.3356	0.8973	0.8956	0.8924	0.9518	0.9673	0.9713
$\geqslant 40\%$	0.3674	0.8642	0.9243	0.8943	0.9645	0.9560	0.9685

TABLE VI
EFFICIENCY ANALYSIS OF DIFFERENT RSIS' SOD METHODS

	CMC [33]	LVNet [9]	PDFNet [10]	SARNet [11]	DAFNet [12]	MJRBM [13]	MFENet
Params (M)	67.83	57.42	78.94	75.62	53.21	48.76	61.57
Speed (FPS)	16	52	48	87	72	84	93

TABLE VII
QUANTITATIVE EVALUATION OF ABLATION STUDY WITH DIFFERENT COMPONENTS ON THE ORSI-4199 DATASET

Baseline	GFP	MFE	SFG	BOM	$F_\beta \uparrow$	$MAE \downarrow$	$S_m \uparrow$
✓					0.8651	0.0263	0.8457
✓	✓				0.8824	0.0192	0.8790
✓	✓	✓			0.9015	0.0132	0.9126
✓	✓	✓	✓		0.9148	0.0109	0.9183
✓	✓	✓	✓	✓	0.9285	0.0084	0.9216

achieves better detection results. For example, in the third image, SARNet [11], PDFNet [10], LVNet [9], and CMC [33] cannot accurately detect the object region and have problems such as error detection and object positioning deviation, while our method realizes accurate detection and positioning of the object region. In addition, the low contrast has a great influence on the detection results. For example, in the ninth image, although the compared methods can detect the

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TABLE VIII
ABLATION STUDY QUANTITATIVE COMPARED RESULTS OF GFP MODULE

	VGG16	V-GFP	ResNet101	R-GFP	DenseNet	D-GFP
$F_\beta \uparrow$	0.6342	0.7583	0.7354	0.7935	0.7562	0.8214
$MAE \downarrow$	0.1572	0.1238	0.1385	0.0983	0.1275	0.0826
$S_m \uparrow$	0.6573	0.7025	0.7283	0.7862	0.7458	0.7935

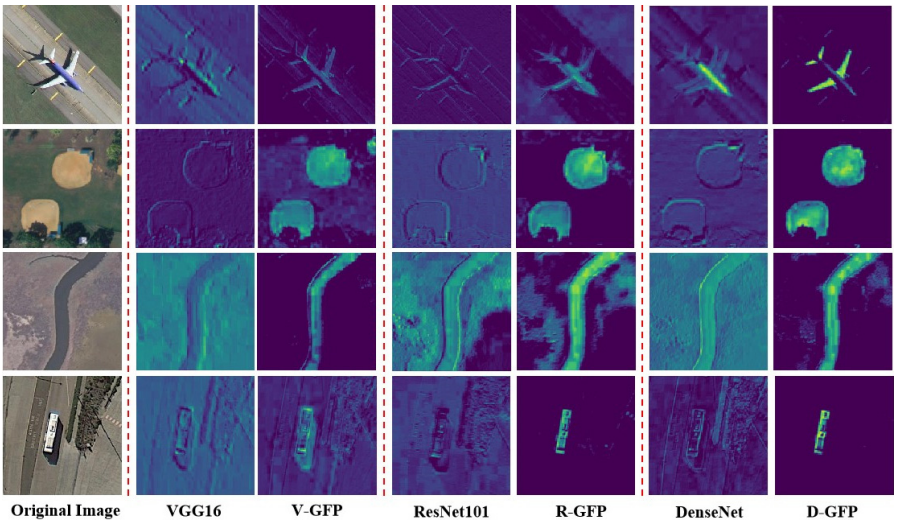


Fig. 13. Feature map visualization of different models. V-GFP (VGG16+GFP), R-GFP (ResNet101 + GFP), and D-GFP (DenseNet + GFP).

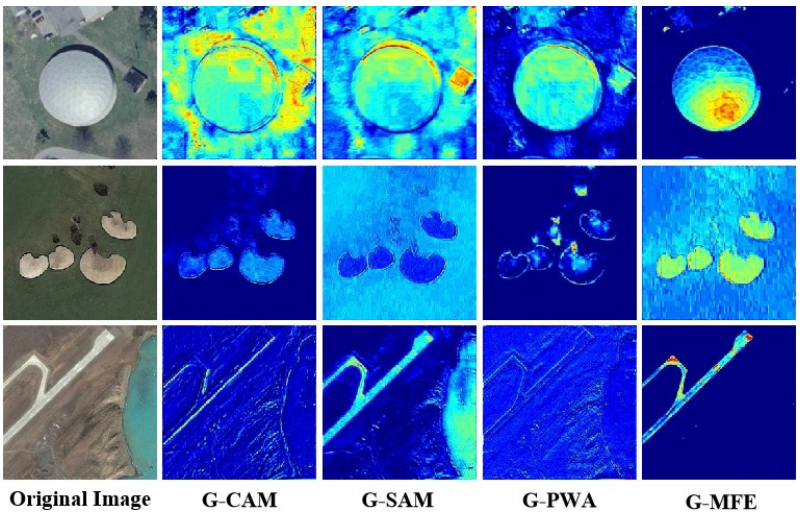


Fig. 14. Feature masks of different attention mechanisms. G-CAM (MFENet+CAM), G-SAM (MFENet + SAM), G-PWA (MFENet + PWA), and G-MFE (MFENet + MFE).

object region, it has an impact on the detection results of MJRBM [13], DAFNet [12], SARNet [11], PDFNet [10], and LVNet [9], while our method still achieved better detection results, which demonstrates that the proposed methods are more robust against imaging interference.

d) Attributes-based study: To understand the characteristics and advantages of our method, we conduct the attribute-based study under different conditions, including scene-, category-, quantity-, and size-based studies. First, in the ORSI-4199 dataset, it contains many typical challenge scenarios, such as big salient object (BSO), SSO, and off center (OC). We compared the proposed MFENet with other methods under different challenge scenarios. To ensure the fairness of comparison, we only use the RSIs SOD method to compare with our method, the performance of different methods under challenge scenarios is shown in Table II. The experimental results of different challenging scenarios show that our method outperforms existing RSIs SOD methods, with an average F-Measure of 0.8831, which is higher than MJRBM and DAFNet by 0.0141 and 0.0234, respectively, and the detection results under different scenarios are optimal. For the category-based study, we conduct the comparative analysis on the ORSSD dataset containing eight different object categories, and the experimental results are shown

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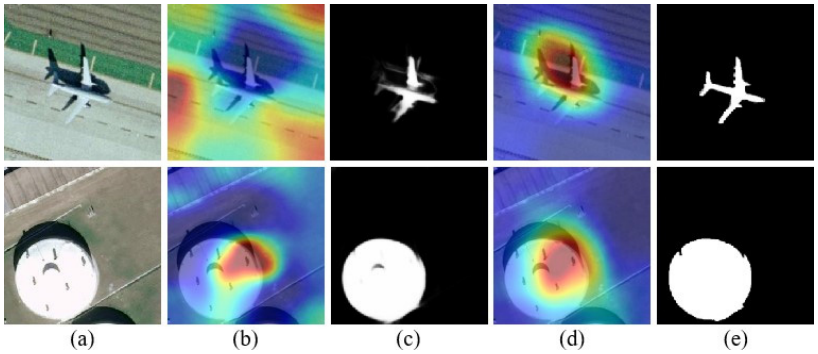


Fig. 15. Visualization of attention feature maps and corresponding salient detection results. (a) Optical RSIs. (b) and (d) Visualization of attention feature maps of MFENet without and with MFE module. (c) and (e) Salient detection results of MFENet without and with MFE module.

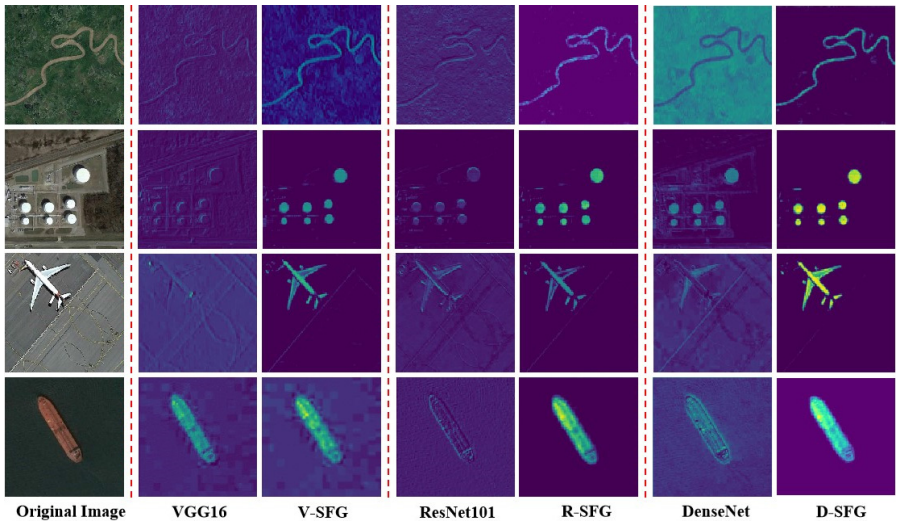


Fig. 16. Feature map visualization of different models. V-GFP (VGG16 + GFP), V-SFG (VGG16 + SFG), R-SFG (ResNet101 + SFG), and D-SFG (DenseNet+SFG).

in Table III. It can be seen from Table III that our method can achieve accurate detection for different category objects, outperforming other compared methods, and the experimental results based on category attributes show that our method has better robustness. For the quantity-based attribute study, we conducted an experimental analysis on the EORSSD dataset containing different quantity targets in each image. It can be seen from the experimental results in Table IV that our method performs the best results and is not affected by the number of targets. For size-based attribute study, which is split based on the proportion of target area in the whole image ($\geqslant$1%, 1%–10%, 10%–20%, 20%–30%, 30%–40%, $\geqslant$40%). From the results in Table V, it can be seen that our method outperforms the other compared methods, and especially for the tiny objects ($\leqslant$1%), it still achieves better detection results.

e) Complexity analysis: To analyze the practicability of different SOD methods, we list the calculation parameters and speed of different RSIs' SOD methods to measure the efficiency. As can be seen from Table VI, the calculation parameters of our method are 61.57M, which is inferior to MJRBM and DAFNet, but it has a faster detection speed, and the FPS reaches 93, which can satisfy the real-time requirements. The large calculation parameters of our method are the use of several different attention mechanisms and MSF branches. Furthermore, the efficiency analysis result shows that our method has improvement space, and it needs to compress the calculation parameters while ensuring detection accuracy.

E. Ablation Study

There are four critical modules within the MFENet, such as the GFP, MFE, SFG, and BOM modules. For the different modules, we conduct ablation experiments to evaluate the necessity and contribution of each module.

The quantitative results of different components are reported in Table VII. The baseline model uses the same encoder–decoder architecture as the GFP module and identical optimization strategies with the full MFENet. Although the baseline model performs poorly in all three metrics, it is still a rather powerful approach compared with other salient detection methods. To obtain the global feature information contained in the optical RSIs, the GFP module is introduced and brings the performance improvement, i.e., the F-measure is improved from 86.51% to 88.24%, the MAE score is decreased from 2.63% to 1.92%, and the S-measure is boosted from 84.57% to 87.90%. Second, to enhance the feature information from the channel and spatial dimensions, the MFE module

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TABLE IX
QUANTITATIVE COMPARISON RESULTS OF SPM WITH DIFFERENT DILATED COEFFICIENTS

(r_1, r_2, r_3)	$F_\beta \uparrow$	$MAE \downarrow$	$S_m \uparrow$
(3, 5, 7)	0.9378	0.0061	0.9254
(3, 7, 9)	0.9285	0.0073	0.9182
(3, 6, 8)	0.9046	0.0085	0.8964
(3, 6, 12)	0.8957	0.0094	0.8850

is used, the F-measure and S-measure are improved from 88.24% and 87.90% to 90.15% and 91.24%, respectively, and the MAE is decreased from 1.92% to 1.32%. Then, we introduced the SFG module to enhance the representation ability of MFENet for semantic features. It can be seen from Table VII that compared with the baseline model, the F-measure and S-measure are increased by 4.97% and 7.26%, respectively, and the MAE is decreased from 2.63% to 1.09%. Finally, to obtain the complete boundary feature, we introduced the BOM module to achieve the construction of our overall model. Compared with different components, the overall architecture of our method achieves the best performance, i.e., the percentage gain reaches 6.34% in terms of F-measure against the baseline model, 7.59% in terms of S-measure, and 1.79% in terms of MAE. The specific advantages of different components are analyzed as follows.

1) GFP Module: The function of GFP is to extract the global feature and detail feature information in RSIs. For comparison, we use VGG16 [53], ResNet101 [54], and DenseNet [55] as baseline models to demonstrate the performance improvement effect of introducing GFP on the MFENet. To output saliency detection results, the baseline model shares other MFENet structures. First, we visualize some feature maps in Fig. 13, it can be seen that compared with the baseline model, the introduction of GFP enables the model to obtain richer feature information and suppress background information interference. Table VIII shows the quantitative comparison results of different strategies. The result shows that the MFENet significantly improves detection accuracy with the introduction of GFP, and the reason is that GFP outperforms other strategies because it has better global feature extraction ability, which can obtain richer feature information in RSIs. In Table IX, we discuss the effect of setting different dilated coefficients r for three parallel dilated convolution layers in the SPM. According to the result comparison, it can be seen that when the dilated coefficient (r_1, r_2, r_3) is set to $(3, 5, 7)$, the model can obtain the optimal results, with F_β, MAE, and S_m reaching 93.78%, 0.61%, and 92.54%, respectively.

2) MFE Module: The purpose of using MFE module is to enhance the model feature extraction ability. Since the essence of MFE is attention mechanism, we use different attention (CAM [39], SAM [43], and positionwise attention (PWA) [56]) to compare with MFE. Fig. 14 shows the feature masks of different attention mechanisms. It can be seen that MFE effectively alleviates the distraction of cluttered backgrounds and focuses on the salient object region. Table X shows the quantitative analysis of different attention mechanisms, where our MFE outperforms other compared attention mechanisms. The reason is that MFE can enhance the feature information from channel dimension and spatial channel so that the model can obtain richer global feature information. To demonstrate that the proposed MFE module can overcome the influence of shadow, we visualize the attention feature map obtained by the MFE module, as shown in Fig. 15. It can be seen that due to the interference of the shadow region, MFENet without the MFE module cannot accurately extract the feature information of the object region. In contrast, MFENet with the MFE module can focus on the feature extraction and representation of the object region, weakening the influence of the shadow region on the salient detection results.

TABLE X
ABLATION STUDY QUANTITATIVE COMPARED RESULTS OF DIFFERENT ATTENTION MECHANISMS

	G-CAM	G-SAM	G-PWA	G-MFE
$F_\beta \uparrow$	0.8247	0.8174	0.8652	0.9435
$MAE \downarrow$	0.0385	0.0247	0.0187	0.0073
$S_m \uparrow$	0.8162	0.8325	0.8716	0.9384

3) SFG Module: The role of SFG is to enable the model to obtain deep semantic feature information contained in RSIs, improving model detection accuracy. To verify the performance of SFG, we continue to use VGG16 [53], ResNet101 [54], and DenseNet [55] as the baseline model, construct VGG16-SFG, ResNet101-SFG, and DenseNet-SFG as the comparison method, and use the BOM to output the detection results. Note that we only use SFG and boundary optimization contained in MFENet. Fig. 16 shows the feature map visualization of different methods, from which it can be seen that the introduction of SFG enables the baseline model to obtain more semantic feature information and suppress background information inference. From Table XI, we can see that the detection accuracy of different baseline models has been significantly improved by using SFG and D-SFG achieving the best detection results. The reason is that DenseNet itself has a deep model structure, which can obtain rich global feature information.

4) BOM Module: To further improve the detection effect of MFENet on the object boundary region, we use the BOM to extract the boundary detail feature information in RSIs. To demonstrate the performance of BOM, we use VGG16 [53], ResNet101 [54], and DenseNet [55] as baseline models, and introduce the BOM to construct VGG16-BOM, ResNet-BOM, and DenseNet-BOM as comparison methods. To intuitively illustrate the effectiveness of the BOM, we provide some visualization detection results in Fig. 17, and it can be seen that the introduction of the BOM can suppress boundary background redundancies and produce more accurate saliency detection results. Table XII shows the quantitative comparison results of different methods, and it can be seen that the use of the BOM can significantly improve the baseline performance, illustrating the effectiveness of the BOM.

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TABLE XI
ABLATION STUDY QUANTITATIVE COMPARED RESULTS OF SFG MODULE

	VGG16	V-SFG	ResNet101	R-SFG	DenseNet	D-SFG
$F_\beta \uparrow$	0.4375	0.7248	0.5214	0.7851	0.8173	0.8315
$MAE \downarrow$	0.1268	0.0842	0.1173	0.0615	0.0942	0.0524
$S_m \uparrow$	0.4512	0.7165	0.5182	0.7732	0.8215	0.8426

TABLE XII
ABLATION STUDY QUANTITATIVE COMPARED RESULTS OF BOM

	VGG16	V-BOM	ResNet101	R-BOM	DenseNet	D-BOM
$F_\beta \uparrow$	0.3024	0.5218	0.4165	0.6483	0.5279	0.7126
$MAE \downarrow$	0.2185	0.1863	0.1538	0.1235	0.1075	0.0865
$S_m \uparrow$	0.3271	0.5324	0.4213	0.6372	0.5381	0.7238

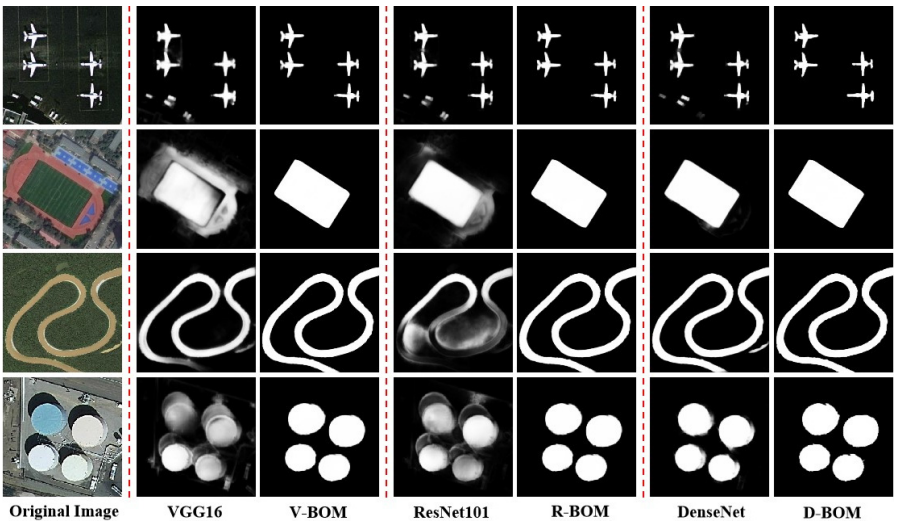


Fig. 17. SOD results of different methods. V-BOM (VGG16+BOM), R-SFG (ResNet101+BOM), and D-SFG (DenseNet + BOM).

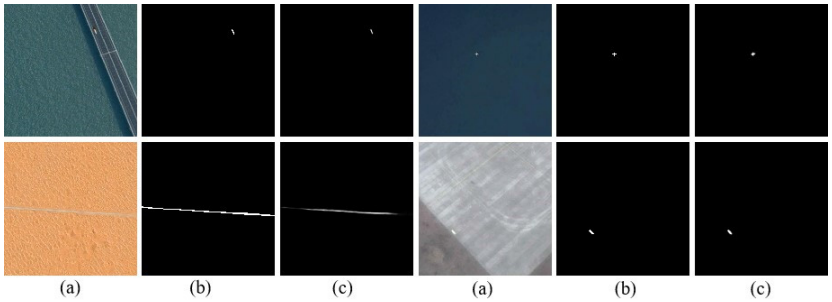


Fig. 18. Failure cases analysis. (a) Optical RSIs. (b) Ground truths. (c) Saliency detection results.

F. Failure Case Analysis

Although the proposed MFENet achieved better salient detection results on several datasets, it still cannot obtain perfect results for very challenging scenarios. In Fig. 18, we show some failure cases of our MFENet.

1) For tiny targets, our method cannot achieve perfect detection results. As shown in the first row of Fig. 18, the proportion of target size in the image is very small, and our method cannot obtain the key feature information so that the detection results have errors with the ground truths. The reason is that to relieve the GPU budget, we cropped and scaled the original image so that the feature of tiny target is lost in the feature extraction.
2) For the target with complex environmental interference, our method fails to achieve accurate saliency detection. As shown in the second row of Fig. 18, the target and background area have high texture similarity, which makes it difficult to accurately detect the target area feature information in the feature extraction process, so our method fails to obtain perfect detection results.

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V. Conclusion

This article focuses on SOD in optical RSIs and proposes a novel SOD framework named MFENet. Inspired by the important role played by the global features, semantic features, and boundary features in SOD, we constructed a GFP module, MFE module, SFG module, and BOM. Specifically, the MFENet first uses the GFP module to extract the global feature in optical RSIs. Then, the MFE module is used to refine the global feature and completes the feature enhancement for tiny object regions. Third, the SFG module is introduced to guide the model to obtain deep semantic feature information. Finally, the BOM enhances the model feature extract ability for salient boundary regions and outputs refined SOD results. Experimental results demonstrate that our proposed MFENet improves the performance on available datasets and outperforms the state-of-the-art methods in different evaluation metrics. In the future, we will introduce unsupervised and semisupervised learning to reduce the dependence on annotation data.

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